ArmorGalv®

NAVSEA
LCAC Test Report

Document #E04

NAVAL SURFACE WARFARE CENTER
PANAMA CITY DIVISION
Panama City, Florida 32407-7001

NSWC PCD

Landing Craft, Air Cushion (LCAC)

Landing Craft, Air Cushion (LCAC) ArmorGalv Gripes Testing, Evaluation and Provisioning

Prepared in response to TPS 11-176

AIR CUSHION VEHICLE BRANCH, Code E32
Naval Surface Warfare Center Panama City Division
Panama City, FL 32407-7001
Tom Whelan
(850) 235-5042
DSN 436-5042

30 SEPTEMBER 2012

DISTRIBUTION STATEMENT D: Distribution authorized to the Department of Defense and U.S. DoD contractors only; administrative/operational use (SEPTEMBER 2012). Other U.S. requests shall be referred to the Commanding Officer, Naval Surface Warfare Center Panama City Division (Code E32), 110 Vernon Avenue, Panama City, FL.

WARNING: This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C. Sec. 2751 et seq.) or the Export Administration Act of 1979, as amended, Title 50, U.S.C., App 2401, et seq. Violations of these export laws are subject to severe criminal penalties. Disseminate per the provisions of OPNAVINST 5510.161.

NOTE: Due to distribution restrictions, we can only share the executive summary.

EXECUTIVE SUMMARY

The Naval Surface Warfare Center Panama City Division (NSWC PCD), Code E32, was tasked by Naval Sea Systems Command (NAVSEA) Program Executive Office, Ships (PMS377) to complete testing, evaluation and provisioning of ArmorGalv Cargo Lashings (Gripes). The ArmorGalv coating is a Zinc-Alloy Thermo Diffusion Coating (TDC) per American Society for Testing and Materials (ASTM) Designation: A 1059A/A 1059M-08. This coating was a by-product of an Office of Naval Research Science and Technology effort. The ArmorGalv coating is far superior to the zinc plating or galvanized coating in use on the legacy Landing Craft, Air Cushion (LCAC) gripes. Testing at NSWC PCD since July 2010 has shown that the ArmorGalv coating will increase the service life of the LCAC gripes and lead to Total Ownership Cost (TOC) reductions.

Assault Craft Unit (ACU)-4 and ACU-5 were outfitted with a ship set (50 each) of ArmorGalv gripes in August 2011 to complete an evaluation on a deployed craft per a NAVSEA approved test plan. ACU-4 outfitted LCAC-84 with the ArmorGalv gripes and deployed aboard the USS *Iwo Jima* in March 2012. ACU-5 deployed LCAC-62 aboard USS *Rushmore* in September 2012 to complete the ArmorGalv gripe evaluation.

Rigorous LCAC gripe pull-testing was conducted through May 2012 at NSWC PCD. The SEA 05 Technical Warrant Holder (TWH) for Machinery – Deck and Underway Replenishment Systems approved the Test Plan, which conducted pull-testing per NAVSEA drawing 804-6397237. The ArmorGalv gripes passed permanent elongation measurement tests, as well as pull to failure tests.

The TWH subsequently approved the use of the Zinc-Alloy TDC (ArmorGalv coating) via email 06 June 2012, as a result of the testing of the coated lashings at NSWC PCD.

Spencer Industries provided an updated LCAC Cargo Lashing drawing, SPN00726, which has been submitted for provisioning purposes. A Temporary National Item Classification Number (T-NICN) of 3990-LL-H97-1943 has been assigned to enable the ACUs to order new ArmorGalv gripes as needed. Since then, a new national stock number (NSN) has also been assigned; it is 3990-01-612-2550. The legacy steel LCAC cargo lashing, NSN 3990-01-506-9254 is now obsolete. Peck & Hale is to provide updated drawing as a result of a meeting held at NSWC PCD on 13 September 2012, so that the ACUs will be able to procure their ArmorGalv coated gripes as well.

ACU-4 post deployment ArmorGalv gripe inspection will be completed at ACU-4 in December 2012 in accordance with Task Planning Sheet (TPS) 13-030. Based on the delay in the ACU-5 deployment and the TWH approval in hand, it is recommended that the ACU-5 post-deployment inspection be deferred.